KNN and K-Mean

1 Introduction

1.1 Project Objective

To construct an automated classification model for banknote authenticity based on geometric features.

1.2 Target Variable

is_genuine (Boolean value indicating genuine or fake)

1.3 Method Selection

The K-Nearest Neighbors (KNN) algorithm was chosen for the following reasons:

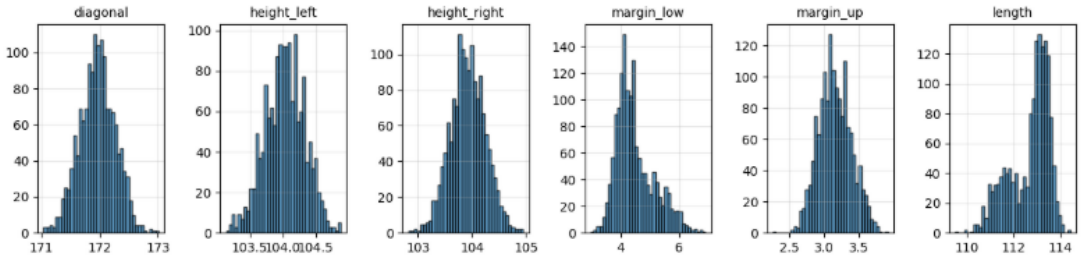
- **Applicability of Principle:** It uses distance measurement to find similar samples, making it a natural fit for classifying items based on their physical characteristics.
- **Data Compatibility:** The dataset is moderately sized (1500 samples) with all features being continuous numerical values.
- **Baseline Value:** Provides a benchmark for subsequent comparison with other algorithms.

Additionally, K-Medoids clustering was applied to explore whether an unsupervised approach could effectively separate genuine and counterfeit banknotes, leveraging its robustness to outliers for this well-structured classification problem.

2 Data Understanding and Preprocessing

2.1 Feature Scale Standardization

- **Problem:** Significant scale differences existed among features (e.g., length ≈ 112mm, margin_up ≈ 3mm). In distance-based algorithms like KNN, this causes features with larger scales to dominate the distance calculation.
- **Method:** Used StandardScaler for feature standardization, transforming all features to a distribution with mean 0 and standard deviation 1.



2.2 Missing Value Handling

- **Problem:** The margin_low feature contained 37 missing values.
- **Method:** Applied mean imputation, filling missing values with the average of the non-missing values for that feature.

2.3 Feature Dimensionality Reduction

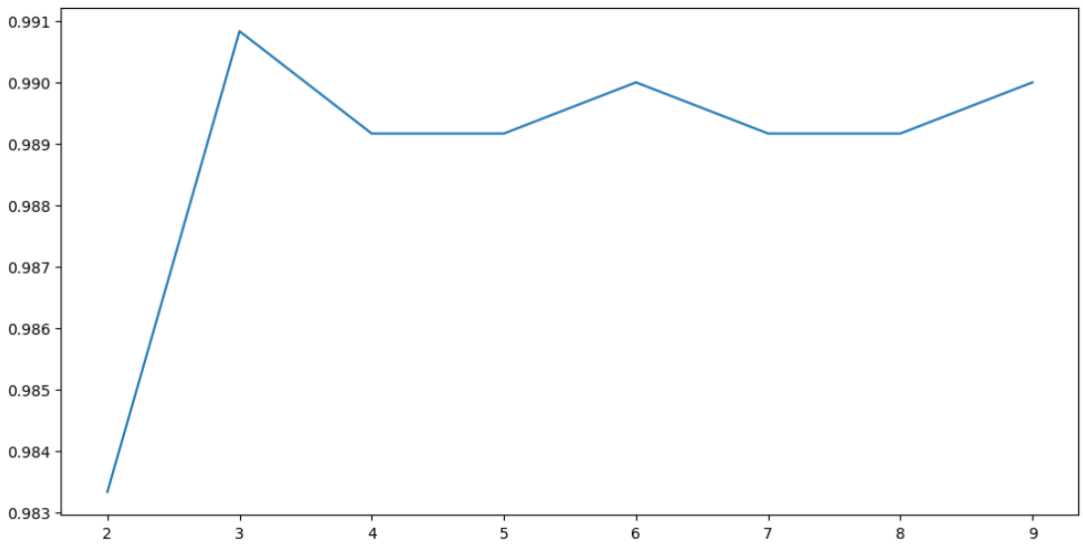
- **Problem:** Correlation analysis revealed information overlap among features (e.g., correlation coefficient between length and is_genuine was 0.85), potentially leading to dimensionality redundancy.
- **Method:** Applied Principal Component Analysis (PCA) with a variance retention threshold of 90%, reducing the feature dimensions from 6 to 5. This preserved the main information while enhancing computational efficiency.

	is_genuine	diagonal	height_left	height_right	margin_low	margin_up	l
is_genuine	1.000000	0.130000	-0.380000	-0.490000	-0.780000	-0.610000	0.8
diagonal	0.130000	1.000000	0.020000	-0.020000	-0.110000	-0.060000	0.1
height_left	-0.380000	0.020000	1.000000	0.240000	0.300000	0.250000	-0.3
height_right	-0.490000	-0.020000	0.240000	1.000000	0.380000	0.310000	-0.4
margin_low	-0.780000	-0.110000	0.300000	0.380000	1.000000	0.430000	-0.6
margin_up	-0.610000	-0.060000	0.250000	0.310000	0.430000	1.000000	-0.5
length	0.850000	0.100000	-0.320000	-0.400000	-0.660000	-0.520000	1.0

3. Model Construction and Optimization

3.1KNN

- **Problem:** The choice of the K value directly impacts model performance—too small leads to overfitting, too large leads to underfitting.
- **Method:** Employed 10-fold cross-validation to systematically evaluate performance for K values ranging from 2 to 9. The performance curve indicated that cross-validation accuracy was highest when K=2, which was therefore determined as the optimal parameter.



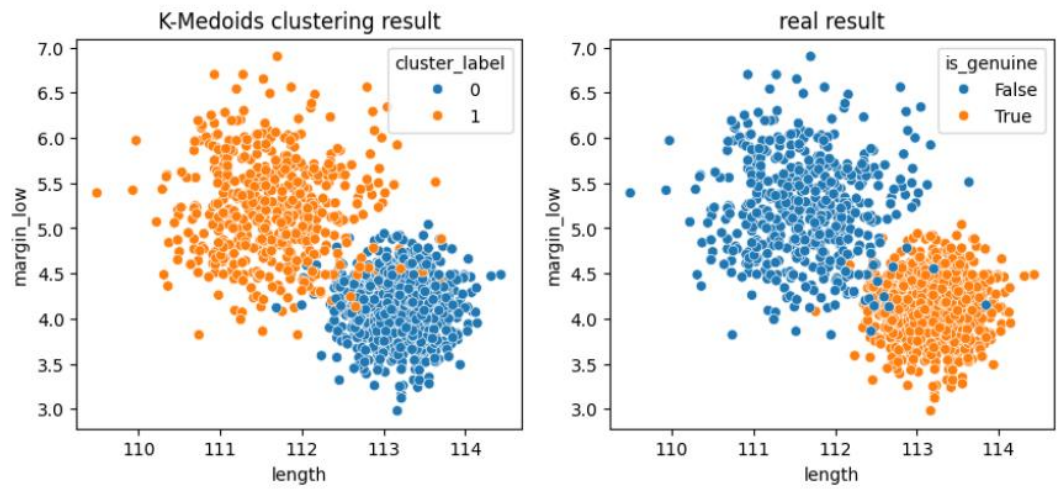
3.2 K-Medoids

- **Problem:** K-Medoids requires selecting the number of clusters (K) and

is sensitive to the choice of distance metric. Unlike supervised KNN, it must discover the inherent grouping structure without using true labels during training.

- **Method:** Applied K-Medoids clustering with Euclidean distance on the standardized feature data. The algorithm was configured with n_clusters=2 to correspond to the genuine/fake categories. A cross-tabulation analysis between cluster labels and true labels (is_genuine) was performed to assess alignment.

- **Result Interpretation:** The clustering results showed strong consistency with true labels, with only a minimal number of misclassifications. This indicates that K-Medoids can effectively separate the two categories in an unsupervised manner, leveraging the geometric feature differences between genuine and counterfeit bills.



4. Performance Comparison and Insights

- **KNN Performance:** Achieved 99% accuracy on the test set with balanced precision and recall across both classes. The model demonstrated excellent generalization capability.
- **K-Medoids Performance:** Achieved accuracy above 98%, with precision and recall metrics exceeding 0.98 for both genuine and counterfeit categories. The scatter plot visualization confirmed that cluster boundaries closely matched the true classification boundaries.

- **Comparative Insight:** Both methods performed exceptionally well, with KNN slightly outperforming K-Medoids in accuracy. This is expected as KNN is a supervised method leveraging labeled data, while

K-Medoids operates unsupervised. However, the strong performance of K-Medoids confirms that the geometric features inherently separate into two distinct clusters, validating the dataset's structure.